I. Singeing, Desizing, Scouring, Bleaching and Mercerization

1. Singeing is carried out to remove:

A. Size from warp yarns

B. Protruding fibres from yarn or fabric

C. Wax and pectin

D. Natural colouring matter

Answer: B) Protruding fibres from yarn or fabric

2. In the common cotton-processing sequence, which process follows singeing?

A. Desizing

B. Mercerization

C. Bleaching

D. Dyeing

Answer: A) Desizing

3. Which of the following is NOT a type of singeing machine?

A. Gas singeing machine

B. Plate singeing machine

C. Roller singeing machine

D. Jigger

Answer: D) Jigger

4. What is the main precaution when singeing fabrics containing synthetic fibres?

A. Fabric must be completely dry

B. Fabric must be passed in rope form

C. Excess heat can melt thermoplastic fibres and form hard beads

D. Alkali must be added

Answer: C) Excess heat can melt thermoplastic fibres and form hard beads

5. Which of the following is NOT a common sizing material?

A. Starch

B. Polyvinyl alcohol (PVA)

C. Carboxymethyl cellulose (CMC)

D. Pectin

Answer: D) Pectin

6. A blue or blue-black colour in the iodine test after desizing indicates:

A. Complete size removal

B. Presence of residual starch

C. Presence of wax

D. Excess alkali

Answer: B) Presence of residual starch

7. The main chemical used for scouring cotton is:

A. Sodium hydroxide (NaOH)

B. Hydrogen peroxide

C. Sodium hypochlorite

D. Sodium chlorite

Answer: A) Sodium hydroxide (NaOH)

8. In bioscouring, which enzymes remove selected impurities under milder conditions?

A. Amylases

B. Cellulases

C. Pectinases

D. Proteases

Answer: C) Pectinases

9. Wool scouring is mainly intended to remove:

A. Sericin gum

B. Wool grease, suint and dirt

C. Starch size

D. Cotton waxes and pectins

Answer: B) Wool grease, suint and dirt

10. Silk degumming is the removal of:

A. Wool grease

B. Suint

C. Starch

D. Sericin gum

Answer: D) Sericin gum

11. Which is the most widely used bleaching agent for cotton?

A. Hydrogen peroxide (H_2O_2)

B. Sodium hypochlorite

C. Sodium chlorite

D. Sodium dithionite

Answer: A) Hydrogen peroxide (H_2O_2)

12. What is the role of a sequestering agent in peroxide bleaching?

A. It increases fabric whiteness by fluorescence

B. It raises the bath temperature

C. It binds metal ions that may catalyse harmful peroxide decomposition

D. It removes starch

Answer: C) It binds metal ions that may catalyse harmful peroxide decomposition

13. Mercerization is the treatment of cotton with:

A. Dilute acid without tension

B. Hydrogen peroxide at the boil

C. Enzymes under mild conditions

D. Concentrated caustic soda, usually under tension

Answer: D) Concentrated caustic soda, usually under tension

14. In a chain mercerizer, the fabric edges are held by:

A. Clips

B. Rollers only

C. Perforated beams

D. Nozzles

Answer: A) Clips

15. Why is tension important in mercerization?

A. It removes size

B. It destroys natural colour

C. It controls shrinkage and improves lustre

D. It increases wax content

Answer: C) It controls shrinkage and improves lustre

16. Optical brightening increases whiteness by:

A. Destroying chromophores by oxidation

B. Increasing apparent whiteness through fluorescence

C. Removing natural waxes

D. Reducing fabric pH

Answer: B) Increasing apparent whiteness through fluorescence

II. Dyes and Their Classification

17. Which statement is correct for direct dyes?

A. They form covalent bonds with cotton

B. They are water-soluble, generally anionic, and salt promotes exhaustion

C. They are insoluble and need reduction

D. They are cationic and used mainly on acrylic

Answer: B) They are water-soluble, generally anionic, and salt promotes exhaustion

18. Reactive dyes are attached to cotton through:

A. Hydrogen bonding only

B. Mechanical entrapment

C. Covalent bond with the fibre

D. Dissolution in the polymer

Answer: C) Covalent bond with the fibre

19. The soluble form of a vat dye applied to the fibre is called the:

A. Leuco dye

B. Dispersed dye

C. Basic dye

D. Metal-complex dye

Answer: A) Leuco dye

20. Acid dyes are suitable for:

A. Cotton, viscose and linen

B. Acrylic only

C. Polyester and acetate

D. Wool, silk and nylon

Answer: D) Wool, silk and nylon

21. The major application of basic (cationic) dyes is:

A. Cotton

B. Acrylic

C. Wool

D. Ordinary polyester

Answer: B) Acrylic

22. The principal dye class for ordinary polyester is:

A. Disperse dyes

B. Direct dyes

C. Acid dyes

D. Vat dyes

Answer: A) Disperse dyes

23. Acid dyes are held on wool mainly by attraction to:

A. Hydroxyl groups of cellulose

B. Hydrophobic polymer chains

C. Protonated amino sites

D. Covalent crosslinks

Answer: C) Protonated amino sites

24. Cationic-dyeable polyester is:

A. Ordinary polyester dyed with acid dyes

B. Polyester dyed only by thermosol

C. Polyester dyed with vat dyes

D. Specially modified polyester that accepts basic dyes

Answer: D) Specially modified polyester that accepts basic dyes

III. Dyeing of Cotton, Silk, Wool, Polyester and Blends

25. In conventional reactive exhaust dyeing of cotton:

A. Alkali promotes exhaustion and salt promotes fixation

B. Both salt and alkali promote only exhaustion

C. Salt promotes exhaustion and alkali promotes fixation

D. Neither salt nor alkali is required

Answer: C) Salt promotes exhaustion and alkali promotes fixation

26. In reactive cold pad-batch dyeing, after padding with dye and alkali the fabric is:

A. Batched for a specified dwell time, then washed and soaped

B. Immediately oxidized

C. Steamed at 180°C

D. Treated with a carrier

Answer: A) Batched for a specified dwell time, then washed and soaped

27. The correct sequence in vat dyeing is:

A. Oxidize → apply → reduce → soap

B. Reduce → apply leuco form → oxidize → soap → rinse

C. Apply → fix with alkali → reduce

D. Soap → reduce → oxidize

Answer: B) Reduce → apply leuco form → oxidize → soap → rinse

28. Typical acid-dye conditions for wool involve:

A. Strongly alkaline bath at room temperature

B. Neutral bath with salt only

C. Dry heat at 200°C

D. Acidic bath with controlled heating, commonly approaching 90–100°C

Answer: D) Acidic bath with controlled heating, commonly approaching 90–100°C

29. HTHP dyeing of polyester is commonly carried out at about:

A. 60–70°C

B. 125–135°C

C. 180–220°C

D. 20–30°C

Answer: B) 125–135°C

30. Thermosol dyeing of polyester is best described as:

A. A carrier method at the boil

B. A cold pad-batch method

C. A continuous pad–dry–dry-heat fixation method at about 180–220°C

D. A low-temperature exhaust method

Answer: C) A continuous pad–dry–dry-heat fixation method at about 180–220°C

31. Reduction clearing after disperse dyeing is used to:

A. Increase dye fixation on cotton

B. Desize the fabric

C. Produce a resist effect

D. Remove surface dye and improve rubbing and wash fastness

Answer: D) Remove surface dye and improve rubbing and wash fastness

32. The common dye combination for a polyester/cotton blend is:

A. Acid + basic

B. Disperse + reactive

C. Basic + vat

D. Acid + direct only

Answer: B) Disperse + reactive

33. Union dyeing means:

A. Dyeing only one fibre of a blend

B. Dyeing components to contrasting colours

C. Dyeing both components to approximately the same shade

D. Leaving one component undyed

Answer: C) Dyeing both components to approximately the same shade

34. Cross dyeing means:

A. Dyeing the components of a blend to different or contrasting colours

B. Dyeing both components to the same shade

C. Leaving one component substantially undyed

D. Dyeing in open width

Answer: A) Dyeing the components of a blend to different or contrasting colours

35. A reserve effect in blend dyeing is one in which:

A. Both components are dyed the same

B. One component remains substantially undyed

C. A resist is printed before dyeing

D. Pigment is fixed by a binder

Answer: B) One component remains substantially undyed

36. Which is a main precaution when dyeing silk?

A. Use strong alkali for fixation

B. Heat rapidly to the boil

C. Use vigorous mechanical agitation

D. Avoid strong alkali, excessive temperature and harsh mechanical treatment

Answer: D) Avoid strong alkali, excessive temperature and harsh mechanical treatment

37. What is the main difficulty in dyeing cotton/wool blends?

A. Cotton's alkaline treatment can damage wool

B. Wool needs thermosol fixation

C. Cotton cannot be dyed with reactive dyes

D. Wool repels all acid dyes

Answer: A) Cotton's alkaline treatment can damage wool

IV. Dyeing Machines

38. In a winch dyeing machine, the fabric:

A. Is wound on a perforated beam

B. Moves in open width between two rollers

C. Moves in rope form as an endless loop

D. Is held stationary while the liquor circulates

Answer: C) Moves in rope form as an endless loop

39. A jigger processes fabric:

A. In rope form

B. In open width, passing between two rollers

C. As loose stock

D. On perforated beams only

Answer: B) In open width, passing between two rollers

40. Which machine uses gentle liquor flow through a nozzle, giving lower tension and abrasion, and is suitable for knits?

A. Soft-flow machine

B. Jigger

C. Winch

D. Padding mangle

Answer: A) Soft-flow machine

41. A limitation of the jigger is:

A. Very high liquor ratio

B. It cannot process woven fabrics

C. It works only above 100°C

D. Higher lengthwise tension with risk of end-to-end variation and creasing

Answer: D) Higher lengthwise tension with risk of end-to-end variation and creasing

42. In beam dyeing:

A. Fabric moves between two rollers

B. Liquor circulates through fabric wound on a perforated beam

C. Fabric moves as an endless rope

D. Dye is transferred from paper

Answer: B) Liquor circulates through fabric wound on a perforated beam

V. Styles of Printing

43. In resist printing, the resist is applied:

A. Before ground dyeing, to prevent colour development locally

B. After ground dyeing, to destroy the ground colour

C. During finishing

D. Only after steaming

Answer: A) Before ground dyeing, to prevent colour development locally

44. Discharge printing involves:

A. Printing on a white ground only

B. Applying wax before dyeing

C. Printing pigment with a binder

D. Destroying an existing ground colour locally, usually activated by steaming

Answer: D) Destroying an existing ground colour locally, usually activated by steaming

45. Wax in batik is an example of:

A. Chemical resist

B. Physical resist

C. Coloured discharge

D. Direct printing

Answer: B) Physical resist

46. In coloured discharge printing:

A. The resisted area remains white

B. The ground is not affected

C. The ground colour is destroyed while a discharge-resistant colour forms the motif

D. Wax prevents dye penetration

Answer: C) The ground colour is destroyed while a discharge-resistant colour forms the motif

VI. Printing Methods

47. In roller printing, the design carrier is:

A. An engraved metal roller

B. A flat mesh screen

C. A perforated nickel cylinder

D. A transfer paper

Answer: A) An engraved metal roller

48. Rotary-screen printing uses:

A. Engraved rollers and doctor blades

B. A cylindrical perforated screen with continuous fabric movement

C. A flat screen with intermittent movement

D. Wax resist

Answer: B) A cylindrical perforated screen with continuous fabric movement

49. Flat-bed screen printing is characterized by:

A. Engraved rollers and continuous movement

B. Cylindrical nickel screens

C. Excellent fine lines from engraving

D. A flat screen and squeegee with usually intermittent fabric movement

Answer: D) A flat screen and squeegee with usually intermittent fabric movement

50. The function of a thickener in a printing paste is to:

A. Provide colour

B. Supply alkali for fixation

C. Control paste flow and maintain design boundaries

D. Cure the binder

Answer: C) Control paste flow and maintain design boundaries

51. Pigment printing normally requires:

A. Binder curing rather than dye–fibre fixation

B. Reduction and oxidation

C. Covalent bonding with cellulose

D. Acidic fixation of leuco dye

Answer: A) Binder curing rather than dye–fibre fixation

VII. Mechanical and Chemical Finishing

52. Calendering is the passing of fabric between rollers under pressure, often with heat, to:

A. Remove size

B. Improve smoothness, lustre, compactness and appearance

C. Bleach the fabric

D. Dye the fabric

Answer: B) Improve smoothness, lustre, compactness and appearance

53. Schreiner calendering produces a silky lustre using:

A. Differential roller speeds

B. Wax-coated rollers

C. Smooth hard rollers only

D. Fine engraved lines on the roller

Answer: D) Fine engraved lines on the roller

54. Compressive shrinking (Sanforizing) is mainly applied to:

A. Wool fabrics

B. Polyester fabrics

C. Woven cotton fabrics

D. Silk fabrics

Answer: C) Woven cotton fabrics

55. Heat setting is used for dimensional stability of:

A. Cotton

B. Thermoplastic fibres such as polyester

C. Wool

D. Silk

Answer: B) Thermoplastic fibres such as polyester

56. The typical sequence in resin (durable-press) finishing is:

A. Pad → Dry → Cure → Afterwash, if required

B. Cure → Pad → Dry

C. Dry → Wash → Pad

D. Steam → Oxidize → Soap

Answer: A) Pad → Dry → Cure → Afterwash, if required

57. A disadvantage of resin finishing is:

A. Increased wrinkling

B. Increased softness

C. Increased tear strength

D. Possible loss of tensile strength, tear strength, abrasion resistance and softness

Answer: D) Possible loss of tensile strength, tear strength, abrasion resistance and softness

58. A water-repellent finish works mainly by:

A. Sealing all pores with a continuous film

B. Increasing fibre-surface energy

C. Lowering fibre-surface energy so water beads and rolls off

D. Dissolving surface waxes

Answer: C) Lowering fibre-surface energy so water beads and rolls off

59. The Limiting Oxygen Index (LOI) is:

A. The minimum oxygen concentration in an oxygen–nitrogen atmosphere needed to sustain burning

B. The maximum nitrogen content in the air

C. The time a fabric burns after ignition

D. The char length after a vertical flame test

Answer: A) The minimum oxygen concentration in an oxygen–nitrogen atmosphere needed to sustain burning

60. Which of the following can be used as antimicrobial finishing agents?

A. Silicone and wax

B. Quaternary ammonium compounds, silver-based agents and chitosan

C. DMDHEU and catalysts

D. Hydrogen peroxide and sodium hydroxide

Answer: B) Quaternary ammonium compounds, silver-based agents and chitosan